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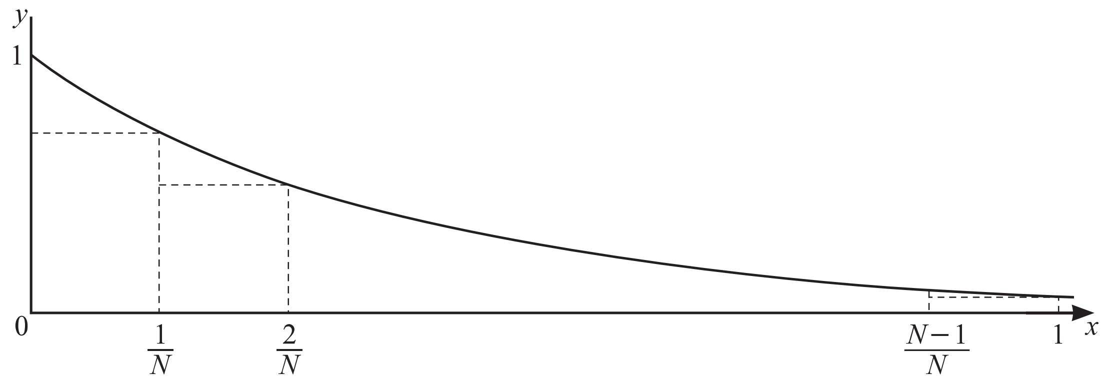

The diagram shows the curve with equation $y = \left(\frac{1}{2}\right)^x$ for $0 \leqslant x \leqslant 1$, together with a set of N rectangles each of width $\frac{1}{N}$.

(a) By considering the sum of the areas of these rectangles, show that $\int_0^1 \left(\frac{1}{2}\right)^x \mathrm{d}x > L_N$, where

$$L_N = \frac{1}{2N\left(2^{\frac{1}{N}} - 1\right)}.$$ [4]

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(b) Use a similar method to find, in terms of N, an upper bound U_N for $\int_0^1 \left(\frac{1}{2}\right)^x \mathrm{d}x$. [4]

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(c) Find the least value of N such that $U_N - L_N \leqslant 10^{-3}$. [2]

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(d) Given that $\int_0^1 \left(\frac{1}{2}\right)^x \mathrm{d}x = \frac{1}{2\ln 2}$, use the value of N found in part **(c)** to find upper and lower bounds for $\ln 2$. [4]

7 (a) Show that an appropriate integrating factor for

$$\sqrt{x^2+16}\,\frac{\mathrm{d}y}{\mathrm{d}x} + y = x\sqrt{x^2+16}$$

is $\frac{1}{4}x + \frac{1}{4}\sqrt{x^2+16}$. [4]